

The system is an old AMD 64 3200+ on an MSI RS480M2 with 1 GB DDR 400 and 80 GB Seagate Barracuda SATA hard disk. My new monitor was a Samsung 740N 17-inch LCD, and I still had my non-functional Samsung 793s CRT lying about. Our first thought was to build the world's first AMD-Mac—something that looked like the older CRT-based Macs, but would run Windows and be AMD-based. A lot of disapproving noises by both Apple fans and those well-versed with legal issues put an end to that thought. The idea stuck though, and we decided we wanted to build a complete system into the CRT monitor casing, using the LCD as the display.

The idea is simple: instead of replacing the CRT on your desk with an LCD, make the CRT casing the entire computer... In what follows, we will help you do this at home.

Stuff You Need

All we could find when rummaging through the office supplies were one metal file, a very flimsy hacksaw, a soldering iron, a screwdriver set, a Swiss army knife, a hand drill, and tonnes of nuts and bolts. We decided to use only these implements, because you should have all this at home, and we wanted to keep this strictly low-budget.

Apart from this, you will need a CRT, preferably non-functional. You can always approach your local computer hardware store and ask him to get you an old 17-inch CRT casing. We enquired, and you can get one for anywhere between Rs 200 and 600, depending on the condition of the case. You will also need an LCD (duh!), your existing PC (which you should completely dismantle before beginning) and a lot of free time and energy.

Breaking It Down

Nimish was the one who volunteered to do all the muscle work (I always knew he was just a destructive little boy at heart), while I was made "Project Manager" by Ed. You have to love all the corporate speak, which means all I am doing is writing this, overseeing the *minions* team, tch-tch-ing away to glory and barking orders! Now this is the life... the ultimate cure for office boredom.

First things first: we had to dismantle the CRT. If you're doing this at home, do so carefully, because CRTs are filled (or is that emptied?) with vacuum. A little heavy jolt or dropping it will result in a rather nasty implosion. The boys dismantled the CRT in the presence of Ajay, our much-hated systems admin—getting hardware out of him is like attaining Nirvana. Very many opened screws later, we picked out the CRT and handed it over to Ajay. Now we finally had what we wanted—the CRT case.

Next, my pretty new LCD had its stand removed. And we found that a 17-inch LCD (non-wide) fits pretty well into a 17-inch flat CRT casing. Just a little cutting and filing required (none by me).

Apparently, a 17-inch CRT has a circuit board holder identical in size to an ATX-size motherboard! All we needed to do was cut the metal container a little on one end to make the back of the motherboard accessible.

We had to get rid of some of the CRT case's internal side supports to get the LCD to fit well.

We used the soldering iron to melt these away. It's not pretty, because it leaves little uneven melted masses, but it does reach those corners easily, and cuts through the plastic casing like butter.

Once we had the LCD seated, it was time to find out just how much screen area was being cut off by the CRT bezel in the front. We found that a good half an inch on all sides was un-viewable, so we drew a few lines and went at it with our little flimsy saw. If you're doing this at home, make sure you don't cut too much off, because you'll never be able to compensate for that. Nimish was a little over-enthusiastic in the beginning, and hacked a little too much away from the top. Our advice: be conservative. It's a lot easier to file a little more of the plastic away than worry about how to cover up faults later. We had already decided we didn't want the LCD's bezel to be visible, as that would spoil the effect of making our mod look almost like a plain old CRT. We cut it accordingly.

This took a whole day to complete, what with all the precise measurements and loads of conservative filing and cutting, after the initial mistakes. If you're doing this alone, it could take considerably longer.

After all this was done, we needed to find a way to fasten the LCD securely in place to the front panel of the CRT case. It's easy to go out and buy something that will do this, but we were determined to use only what we had. Luckily, the old CRT had earthing that ran around the back of the tube, and also acted as a rope that held it in place. Since it was made for this case, it fit perfectly. It even came with a spring, so you can pull it as tight as you require. We screwed in four screws at each corner and literally tied the LCD to the front panel.

Our very next challenge was to make the LCD's controls accessible. Our CRT had its buttons on the right, while the LCD had its buttons on the front. Quite a dilemma! We then decided that we could use the little holes of the CRT's buttons as front panel USB and audio, and decided to just cut away the CRT's plastic bezel to reveal the LCD's controls. If you're lucky enough to have both an LCD and a CRT with approximately the same button placement, you can rig the same holes in the CRT's casing to be used.

We put the power/reset and HDD/power LEDs on the other end. Just a little cutting and *melting* with a soldering iron is required to achieve this.

This completed the front panel, which now held the LCD monitor, front panel audio and USB, and the power switches.

Component Input Time

The next thing we had to do was fit the motherboard into where the CRT's circuit board once was. Now the problem with the circuit board's holder was that it's made of metal.

One very important thing to remember when fixing your motherboard into a case-modded system is insulation. Never seat your motherboard flush against a metal back. If you look carefully into your cabinet, you will see that the motherboard actually sits a little higher than the back on which it rests. This is done using little elevated screws, which prevent any of the soldered-on contacts behind the motherboard from